Chapter 10

Flow-Field and Force Measurement

This chapter will examine the topic of aerodynamic force measurement in solid-wall and open-jet wind tunnels. Both direct force measurement (near-field) and wake survey (far-field) methods will be considered.

10.1 Wind Tunnel Methods – Overview

The two most common purposes of wind tunnel tests are:

1. To determine what the forces and moments would be on the actual object in flight. The tests can be performed either on the actual object, or more commonly on a suitable scale model.
2. To obtain data for validation or calibration of theoretical or computational methods, using some representative model geometry and flow conditions. In addition to the overall loads, ideally this data includes more detailed information such as surface pressures and flow-field velocity distributions.

If the intent is to simulate an unbounded flow-field about a 2D airfoil or a 3D aircraft in flight, then the tunnel walls or tunnel jet boundaries will have unwanted influences on the flow velocities at the model location and hence on the model aerodynamic loads. Modeling and estimation of these tunnel boundary effects to correct measured data is a major goal of this chapter. Another goal is to give a sense of the theoretical and practical challenges in wind tunnel measurements. For a much more complete treatment of wind tunnel testing methods and techniques, see Barlow et al [67].

10.2 Direct Force Measurements

This section examines the measurement of the aerodynamic forces and moments acting on a model mounted in a wind tunnel, using a load sensor.

10.2.1 Force component definitions and rotations

The angle of attack α is generally defined between the model reference line and the direction of the *tunnel freestream*, which is defined as the velocity vector which is present in the empty tunnel. It will be denoted here by $\mathbf{V}_\infty$, even though there is no flow "at infinity" in a wind tunnel. Assuming the tunnel freestream is horizontal, the axis for the lift force component L is most conveniently chosen to be vertical, along the gravity vector. The axis for the sideforce Y is then defined orthogonal to the other two.

The force vector acting on a wind tunnel model can be measured by mounting the model on a load sensor. After the model weight and other instrumentation biases (or "wind-off tares") are properly subtracted, the remaining force $\mathbf{F}$ consists of the integrated pressure and shear forces acting on the model surface, and hence is the aerodynamic near-field force as defined in Chapter 5. This force is reported as components along the sensor's *sensing axes*.

For a sensor which is solidly attached to the model and moves with it, it is most convenient to define the model's xyz geometry axes to lie along the sensor axes, so that the load sensor directly reports F_x, F_y, F_z, as shown in Figure 10.1. These forces must then be projected onto the freestream axes to give the lift L and drag D, by using the overall model angle of attack α and the stability-axis matrix relation (6.5), restated as follows.

$$\begin{aligned} D &= F_x \cos\alpha + F_z \sin\alpha \\ Y &= F_y \\ L &= F_z \cos\alpha - F_x \sin\alpha \end{aligned} \tag{10.1}$$

If the model is also at a sideslip angle then the wind-axis matrix relation (6.6) would be used instead. This more general case will not be considered here.

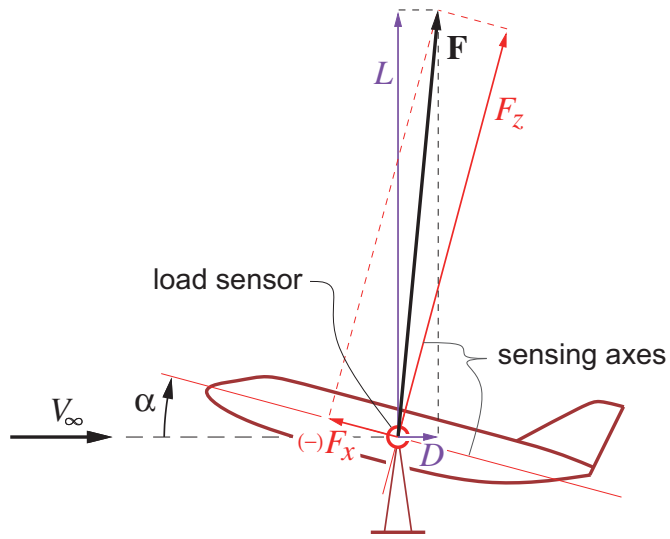

Figure 10.1: Axial and normal force components F_x, F_z along load sensor axes resolved into drag and lift components D, L along and normal to the freestream. Axial force F_x is defined positive towards the tail, opposite to direction shown.

In some wind tunnel installations the load sensor is fixed relative to the wind tunnel rather than the model, and the model then pivots on the sensor to change the angle of attack. Ideally this type of sensor is fixed with its axes aligned with the freestream so that it directly reports the D, Y, L force components, and the rotation calculations (10.1) are then skipped.

10.2.2 Drag measurement error sensitivity

The measurement of aerodynamic forces by a load sensor generally gives accurate lift and sideforce components. However, if the lift and sideforce are very large compared to the drag, as is typical for most aircraft, then measuring the drag accurately is much more difficult. A major potential source of error is misalignment of the sensor relative to its assumed orientation. This is sketched in Figure 10.2, for the case $\alpha = 0$ for simplicity. If there is a small alignment error angle θ_{err}, as in Figure 10.2 on the right, the sensor will report the following incorrect forces $L_{\text{err}}, D_{\text{err}}$, and corresponding fractional drag error.

$$L_{\text{err}} = L \cos\theta_{\text{err}} - D \sin\theta_{\text{err}} \simeq L \tag{10.2}$$

$$D_{\text{err}} = D \cos\theta_{\text{err}} + L \sin\theta_{\text{err}} \simeq D + L\,\theta_{\text{err}} \tag{10.3}$$

$$\frac{D_{\text{err}} - D}{D} = \frac{L}{D}\,\theta_{\text{err}} \tag{10.4}$$

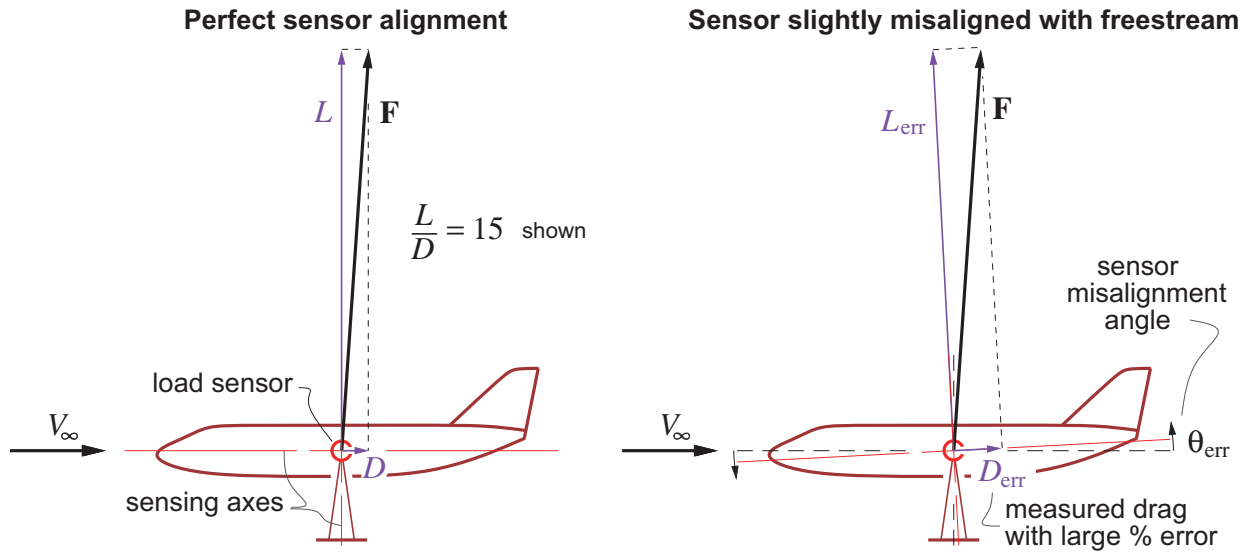

Figure 10.2: Ideal situation for measuring aerodynamic forces with a load sensor (left). In the presence of lift, a small misalignment of sensing axes from freestream direction results in a large fractional error in the measured drag (right).

In the usual situation which has $L \gg D$, the $D \sin\theta_{\rm err}$ lift term is negligible so the effect of $\theta_{\rm err}$ on the measured lift is very small. But the $L \sin\theta_{\rm err}$ drag term is very significant and can cause large fractional measured drag errors. For example, for the case sketched in Figure 10.2 which has $L/D = 15$, an axis misalignment of $\theta_{\rm err} = 1^\circ = 0.01745\,{\rm rad}$ will produce a very large 26% error in the measured drag.

The angle error can be due to either an incorrect alignment of the sensor with the model, or an incorrect measurement of the angle of attack, or possibly due to the actual tunnel freestream flow direction being misaligned with the assumed direction. Obtaining small drag measurement errors for a lifting model therefore as a minimum requires very precise geometric and aerodynamic angle measurements.

For the remainder of this chapter we will assume that sufficient care has been used with the angle measurements and load sensor alignment so that the true force and moment components on the model are correctly measured. We will also assume that the tunnel freestream dynamic pressure q_∞ corresponding to V_∞ is correctly measured, sufficiently far from the model to avoid any influence from the model's near-field flow.

10.2.3 Uncorrected coefficients

Even with perfect measurement accuracy, measured loads which are intended to represent unbounded flow are still considered to be *uncorrected* for wind tunnel wall or jet boundary effects, and are denoted by the $(\,)_{\rm u}$ subscript. However, they can still be non-dimensionalized by the freestream dynamic pressure and the chosen geometric reference quantities into their corresponding coefficients. Corrections for both 2D and 3D coefficients will be derived, but for brevity only the lift, drag, and pitching moment will be presented.

$$c_{\ell_{\rm u}} \equiv \frac{L'_{\rm u}}{q_\infty c_{\rm ref}} \quad , \quad c_{d_{\rm u}} \equiv \frac{D'_{\rm u}}{q_\infty c_{\rm ref}} \quad , \quad c_{m_{\rm u}} \equiv \frac{\mathcal{M}'_{\rm u}}{q_\infty c_{\rm ref}^2} \tag{10.5}$$

$$C_{L_{\rm u}} \equiv \frac{L_{\rm u}}{q_\infty S_{\rm ref}} \quad , \quad C_{D_{\rm u}} \equiv \frac{D_{\rm u}}{q_\infty S_{\rm ref}} \quad , \quad C_{m_{\rm u}} \equiv \frac{\mathcal{M}_{\rm u}}{q_\infty S_{\rm ref} c_{\rm ref}} \tag{10.6}$$

The uncorrected angle of attack $\alpha_{\rm u}$ is the geometric model angle, and the uncorrected Reynolds number is $Re_{\rm u} = V_\infty c_{\rm ref}/\nu$. The subsequent sections will address the problem of how to use this uncorrected data to obtain the *corrected* coefficients which would be obtained from the model if it was operated in an unbounded flow at the corrected angle of attack and corrected Reynolds number.

10.3 Wind Tunnel Corrections

The topic of wind tunnel corrections is a fairly large field, which started with the work of Glauert [68]. Later work by Allen and Vincenti [69] considered compressible flows, and many other refinements have been developed since. The objective here is to formulate the wind tunnel corrections using the flow-field modeling concepts and far-field models developed in Chapter 2. This will further illustrate the effectiveness of these concepts, and will also provide readily accessible means for computing the corrections in applications.

The approach here follows the classical method of images to represent the influence of the walls or jet boundaries in the wind tunnel. After the image strengths are set by the wall or jet boundary conditions, they are then used to give velocity perturbations $\Delta u(x)$ and $\Delta w(x)$ along the tunnel axis. These velocity perturbations in turn produce changes in the model's flow-field and forces, which will be estimated using the various aerodynamics models and methods developed throughout this book. Removal of these predicted changes from the measured data and estimating their effects on the flow conditions constitutes wind tunnel corrections.

To allow focusing on the concepts rather than the details, the tunnel correction formulation and procedure will first be given for the 2D case, for both solid-wall and open-jet tunnels. The 3D solid-wall and open-jet tunnel cases will then be examined.

10.3.1 2D solid-wall boundaries

A solid-wall wind tunnel has the flow-tangency condition $w = 0$ everywhere on its top and bottom walls. This flow can be constructed by adding a cascade of +/- images to the real body's source and vortex sheets, as shown in Figure 10.3. The source sheets of the images have the same sign as the sheets representing the real model, while the vortex sheets of the images alternate in sign. In this superposition, $\mathbf{V}_\infty$ is the velocity in the tunnel with the model (and hence the images) absent, consistent with the previous definition.

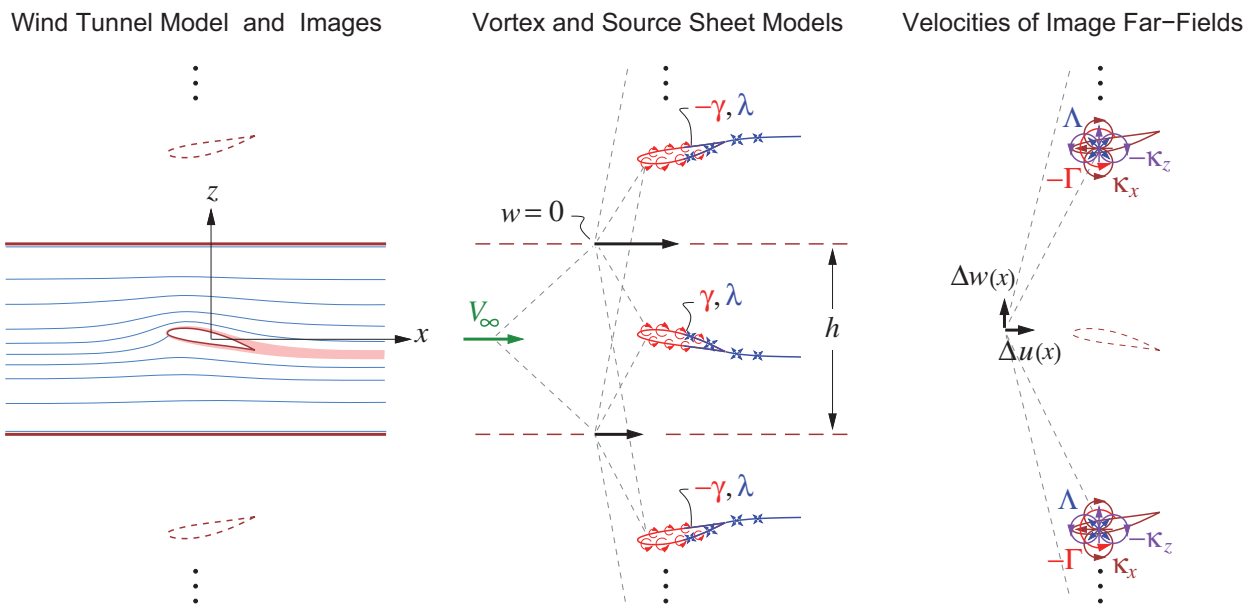

Figure 10.3: 2D airfoil model in tunnel (left) is modeled as an infinite cascade of +/- vortex-sheet and source-sheet images which give physically-correct straight streamlines at the wall locations (center). These images have equivalent lumped 2D point singularities (right). Images produce velocity changes $\Delta u(x), \Delta w(x)$ along the centerline which would be absent in unbounded flow.

Next, we sum the horizontal and vertical velocity contributions $\Delta u(x), \Delta w(x)$ of the images alone, at

some location x along the tunnel centerline. The far-field model (2.79) is used for each image, so that these velocities can be expressed in terms of independent unit-strength components (or kernel functions) $\hat{u}_\Lambda, \hat{u}_{\kappa_x}, \hat{w}_\Gamma, \hat{w}_{\kappa_z}$ which are summations over all the images' far-field sources, vortices, and doublets. The normalized tunnel coordinates X, Y, Z will be convenient here.

$$\Delta u(x) = \frac{\Lambda}{h}\,\hat{u}_\Lambda + \frac{\kappa_x}{h^2}\,\hat{u}_{\kappa_x} \tag{10.7}$$

$$\Delta w(x) = \frac{\Gamma}{h}\,\hat{w}_\Gamma + \frac{\kappa_z}{h^2}\,\hat{w}_{\kappa_z} \tag{10.8}$$

$$\hat{u}_\Lambda(x) = \frac{1}{\pi}\left[\frac{X}{1+X^2} + \frac{X}{4+X^2} + \frac{X}{9+X^2} + \ldots\right] + \frac{1}{2} \qquad \text{(2D solid walls)} \tag{10.9}$$

$$\hat{w}_\Gamma(x) = \frac{1}{\pi}\left[\frac{X}{1+X^2} - \frac{X}{4+X^2} + \frac{X}{9+X^2} - \ldots\right] \qquad \text{(2D solid walls)} \tag{10.10}$$

$$\hat{u}_{\kappa_x}(x) = \frac{1}{\pi}\left[\frac{1-X^2}{(1+X^2)^2} + \frac{4-X^2}{(4+X^2)^2} + \frac{9-X^2}{(9+X^2)^2} + \ldots\right] \qquad \text{(2D solid walls)} \tag{10.11}$$

$$\hat{w}_{\kappa_z}(x) = \frac{1}{\pi}\left[\frac{1-X^2}{(1+X^2)^2} - \frac{4-X^2}{(4+X^2)^2} + \frac{9-X^2}{(9+X^2)^2} - \ldots\right] \qquad \text{(2D solid walls)} \tag{10.12}$$

$$X \equiv \frac{x}{h} \quad , \qquad Y \equiv \frac{y}{h} \quad , \qquad Z \equiv \frac{z}{h} \tag{10.13}$$

Each term in the brackets is the contribution of one image pair above and below the tunnel. The added $1/2$ term in (10.9) is needed to obtain $\Delta u = 0$ for large negative X at the tunnel inflow. The additional useful relations

$$\frac{\mathrm{d}\hat{u}_\Lambda}{\mathrm{d}X} = \hat{u}_{\kappa_x} \quad , \qquad \frac{\mathrm{d}\hat{w}_\Gamma}{\mathrm{d}X} = \hat{w}_{\kappa_z} \tag{10.14}$$

will also be used to formulate pressure-gradient and flow-curvature corrections.

For any X value the summations can be performed numerically with as many terms as needed to obtain the asymptotic result to a sufficient tolerance. However, at the model location $X = 0$, the sums have the following known analytic results.

$$\hat{u}_\Lambda(0) = \frac{1}{2} \ , \quad \hat{w}_\Gamma(0) = 0 \ , \quad \hat{u}_{\kappa_x}(0) = \frac{\pi}{6} \ , \quad \hat{w}_{\kappa_z}(0) = \frac{\pi}{12} \qquad \text{(2D solid walls)} \tag{10.15}$$

The $\Delta u, \Delta w$ superpositions defined by equations (10.7)–(10.12) correspond to z locations midway between the top and bottom walls, which is where the model is assumed to be positioned. Away from the centerline Δu would also have terms with Γ and κ_z contributions, and Δw would have terms with Λ and κ_x contributions. To get some insight into the nature of wind tunnel wall effects, the complete 2D superpositions were used to compute streamline patterns of the "effective freestream" velocity field

$$\mathbf{V}_{\text{eff}}(x,z) = \mathbf{V}_\infty + \Delta u(x,z)\,\hat{\mathbf{x}} + \Delta w(x,z)\,\hat{\mathbf{z}} \tag{10.16}$$

for a number of basic wind tunnel models having only one far-field component present. The resulting streamline patterns are shown in Figures 10.4 and 10.5, one component at a time. Four basic effects can be identified. These are discussed and quantified next.

Volume and wake blockage

The model volume, or cross-sectional area A in 2D, produces a local constriction of the $\mathbf{V}_{\text{eff}}$ streamlines via the image doublets κ_x. The model's drag and associated wake displacement thickness δ^*_∞ produces a further constriction via the image sources Λ. Both velocity perturbations are shown in Figure 10.6. Note that the

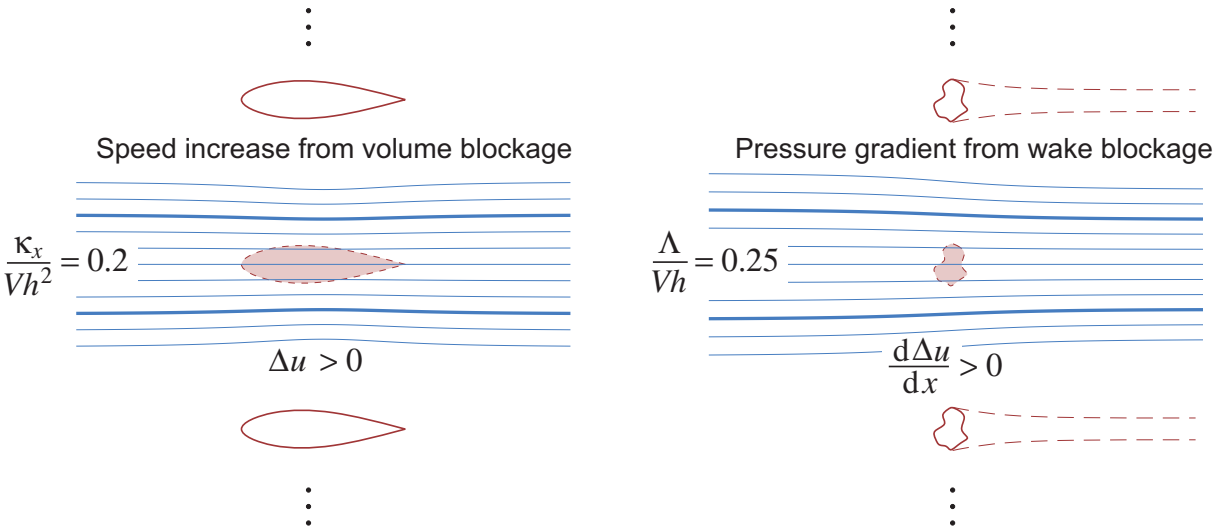

Figure 10.4: Streamlines of the effective freestream $\mathbf{V}_{\text{eff}}(x,z)$ resulting from model images required to model effects of solid tunnel walls. Shown are models with only volume (left) and only profile drag (right). Images have a positive Δu and $\mathrm{d}\Delta u/\mathrm{d}x$ along the tunnel centerline at the real model location.

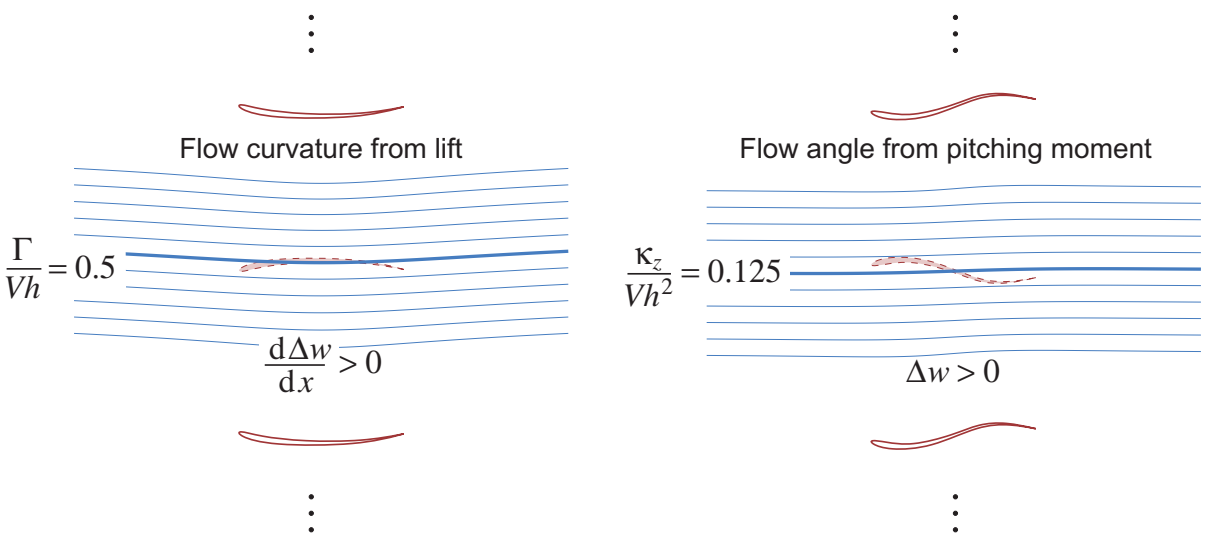

Figure 10.5: Streamlines of the effective freestream $\mathbf{V}_{\text{eff}}(x,z)$ resulting from model images required to model effects of solid tunnel walls. Shown are models with only lift (left) and only pitching moment (right). Images have a positive $\mathrm{d}\Delta w/\mathrm{d}x$ and Δw along the tunnel centerline at the real model location.

doublet effect is local while the source effect persists downstream. At the model at $x=0$, these perturbations due to the volume and the wake displacement produce an increase ΔV in the effective freestream speed, known as the *blockage effect*. To first order this depends only on the streamwise velocity perturbation Δu.

$$V_{\text{eff}} = V_\infty + \Delta V \tag{10.17}$$

$$\frac{\Delta V}{V_\infty} = \frac{\Delta u(0)}{V_\infty} = \frac{1}{2}\frac{\Lambda}{V_\infty h} + \frac{\pi}{6}\frac{\kappa_x}{V_\infty h^2} = \frac{1}{4}\frac{c}{h}c_d + \frac{\pi}{6}\frac{A}{h^2} \qquad \text{(2D solid walls)} \tag{10.18}$$

The far-field source and doublet relations (2.85) and (2.89) have been used to write Λ and κ_x in terms of the model's drag coefficient c_d and cross-sectional area A, with the assumption $V_{\text{eff}} \simeq V_\infty$. This is justified

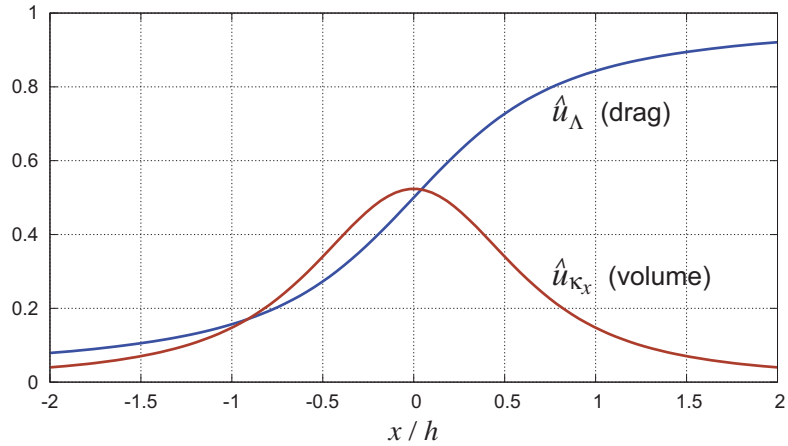

Figure 10.6: Streamwise velocity disturbances along the tunnel centerline due to solid walls, for unit-strength far-field doublet (from model volume) and far-field source (from model drag).

because $\Delta V/V_\infty$ is a small quantity, so this will incur only a second-order error which is negligible.

The small increase in the effective freestream speed also produces the following increased effective freestream dynamic pressure at the model location.

$$q_{\text{eff}} \equiv \frac{1}{2}\rho V_{\text{eff}}^2 = q_\infty\left(1+\frac{\Delta V}{V_\infty}\right)^2 \simeq q_\infty\left(1+2\frac{\Delta V}{V_\infty}\right) = q_\infty\left(1+\frac{1}{2}\frac{c}{h}c_d+\frac{\pi}{3}\frac{A}{h^2}\right) \quad (10.19)$$

It will also be useful to approximate the inverse dynamic pressure ratio

$$\frac{q_\infty}{q_{\text{eff}}} = \left(1+\frac{1}{2}\frac{c}{h}c_d+\frac{\pi}{3}\frac{A}{h^2}\right)^{-1} \simeq 1-\frac{1}{2}\frac{c}{h}c_d-\frac{\pi}{3}\frac{A}{h^2} \qquad \text{(2D solid walls)} \quad (10.20)$$

which assumes that the two correction factors are much less than unity.

Wake-induced buoyancy

An additional effect of the wake displacement is that it produces an acceleration of the effective freestream at the model location $x=0$, as can be seen from the $\hat{u}_{\Lambda(x)}$ curve in Figure 10.6.

$$\frac{\mathrm{d}V_{\text{eff}}}{\mathrm{d}x} = \frac{\mathrm{d}\Delta V}{\mathrm{d}x} = \frac{\mathrm{d}\Delta u}{\mathrm{d}x}(0) = \frac{\Lambda}{h^2}\frac{\mathrm{d}\hat{u}_\Lambda}{\mathrm{d}X}(0) = \frac{\pi}{6}\frac{\Lambda}{h^2} = \frac{\pi}{12}\frac{V_\infty c}{h^2}c_d \quad (10.21)$$

The result is a streamwise pressure gradient

$$\frac{\mathrm{d}p_{\text{eff}}}{\mathrm{d}x} = -\rho V_\infty\frac{\mathrm{d}V_{\text{eff}}}{\mathrm{d}x} = -\frac{\pi}{12}\frac{\rho V_\infty^2 c}{h^2}c_d \quad (10.22)$$

which then acts on the airfoil's cross-sectional area to produce an added *buoyancy drag* which will need to be subtracted from the uncorrected drag.

$$\Delta D'_{\text{buoy}} = -\frac{\mathrm{d}p_{\text{eff}}}{\mathrm{d}x}A = \frac{\pi}{12}\rho V_\infty^2 c\, c_d\frac{A}{h^2} \quad (10.23)$$

$$\Delta c_{d_{\text{buoy}}} = \frac{D'_{\text{buoy}}}{\frac{1}{2}\rho V_\infty^2 c} = \frac{\pi}{6}\frac{A}{h^2}c_d \qquad \text{(2D solid walls)} \quad (10.24)$$

Streamline curvature

The model lift and associated image circulations produce a local flow curvature χ via the streamwise derivative $\mathrm{d}\Delta w/\mathrm{d}x$,

$$\chi \;=\; \frac{1}{V_\infty}\frac{\mathrm{d}\Delta w}{\mathrm{d}x}(0) \;=\; \frac{\Gamma}{V_\infty h^2}\frac{\mathrm{d}\hat{w}_\Gamma}{\mathrm{d}X}(0) \;=\; \frac{\pi}{12}\,\frac{\Gamma}{V_\infty h^2} \;=\; \frac{\pi}{24}\,\frac{c}{h^2}\,c_{\ell_\mathrm{u}} \tag{10.25}$$

where the far-field vortex relation (2.84) has been used. Again, the assumption $V_\mathrm{eff} \simeq V_\infty$ is made here.

This flow curvature effectively adds to the airfoil's real curvature and thus changes its lift and pitching moment. The effects can be estimated by first-order thin airfoil theory as given in Appendix D, by specifying a parabolic camberline with the specified curvature. This produces the following change in the lift and moment coefficients which will need to be subtracted from the uncorrected values.

$$\Delta c_{\ell_\mathrm{curv}} \;=\; \frac{\pi}{2}\chi\, c \;=\; \frac{\pi^2}{48}\,\frac{c^2}{h^2}\,c_{\ell_\mathrm{u}} \qquad \text{(2D solid walls)} \tag{10.26}$$

$$\Delta c_{m_\mathrm{curv}} \;=\; -\frac{\pi}{8}\chi\, c \;=\; -\frac{\pi^2}{192}\,\frac{c^2}{h^2}\,c_{\ell_\mathrm{u}} \qquad \text{(2D solid walls)} \tag{10.27}$$

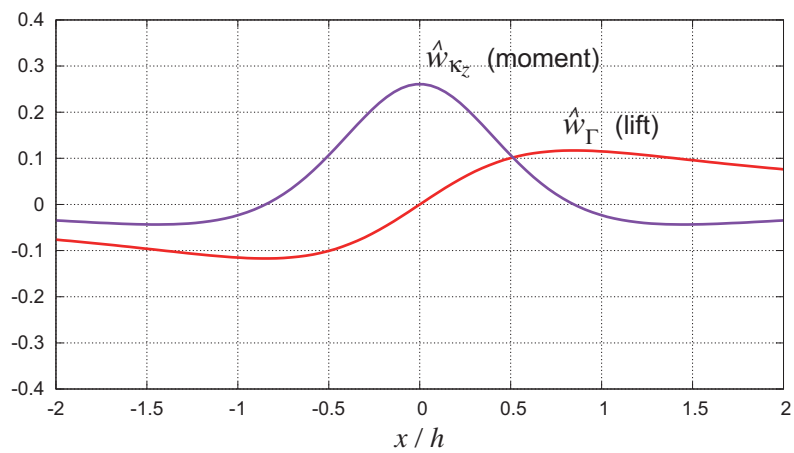

Figure 10.7: Vertical velocity disturbances along the tunnel centerline due to solid walls, for unit-strength far-field vortex (from model lift) and far-field z-doublet (from model pitching moment).

Flow angle

The far-field z-doublet is given by the pitching moment about the far-field singularity location, which is somewhat arbitrary. However, the most rational choice is to choose this location at the lift centroid, which is also the vorticity centroid. In that case the pitching moment about this location and the z-doublet are both zero, so no lift correction needs to be applied for the $\hat{w}_{\kappa_z}$ perturbation velocity component.

$$\Delta c_{d_\mathrm{angle}} \;=\; 0 \tag{10.28}$$

However, the flow angle perturbation represents a direct correction to the experimental angle of attack.

$$\begin{aligned}\Delta\alpha_\mathrm{angle} \;&=\; \frac{\Delta w(0)}{V_\infty} \;=\; \frac{\kappa_z}{V_\infty h^2}\,\hat{w}_{\kappa_z}(0) \;=\; \frac{\pi}{3}\frac{c^2}{h^2}\,(c_{m_\mathrm{u}})_{c/2} \\ &=\; \frac{\pi}{3}\frac{c^2}{h^2}\left(c_{m_\mathrm{u}} + \frac{1}{4}\,c_{\ell_\mathrm{u}}\right) \qquad \text{(2D solid walls)}\end{aligned} \tag{10.29}$$

The midchord pitching moment coefficient $(c_{m_\mathrm{u}})_{c/2}$ was used to define the far-field z–doublet, since that's where the far-field vortex was assumed to be placed for the flow-curvature corrections above.

Application of 2D wall corrections

We now assume that all the velocity perturbations from the images are small, so that the resulting force perturbation from each velocity perturbation can be subtracted linearly from the uncorrected coefficients. The dynamic pressure scaling is then applied to produce the final corrected coefficients.

$$c_d = \frac{q_\infty}{q_{\text{eff}}}\left(c_{d_{\text{u}}} - \Delta c_{d_{\text{buoy}}} - \Delta c_{d_{\text{angle}}}\right) \tag{10.30}$$

$$c_\ell = \frac{q_\infty}{q_{\text{eff}}}\left(c_{\ell_{\text{u}}} - \Delta c_{\ell_{\text{curv}}}\right) \tag{10.31}$$

$$c_m = \frac{q_\infty}{q_{\text{eff}}}\left(c_{m_{\text{u}}} - \Delta c_{m_{\text{curv}}}\right) \tag{10.32}$$

The angle of attack and Reynolds number which the model effectively sees are also obtained by correcting for the perturbed flow angle and flow speed.

$$\alpha = \alpha_{\text{u}} + \Delta\alpha_{\text{angle}} \tag{10.33}$$

$$Re = \frac{V_{\text{eff}}}{V_\infty} Re_{\text{u}} \tag{10.34}$$

Substituting the various changes into the correction relations derived from the solid-wall image models and dropping the higher-order terms, gives the corrected coefficients in terms of the uncorrected values.

$$c_d = \left(1 - \frac{1}{2}\frac{c}{h}c_d - \frac{\pi}{2}\frac{A}{h^2}\right)c_{d_{\text{u}}} \tag{10.35}$$

$$c_\ell = \left(1 - \frac{1}{2}\frac{c}{h}c_d - \frac{\pi}{3}\frac{A}{h^2} - \frac{\pi^2}{48}\frac{c^2}{h^2}\right)c_{\ell_{\text{u}}} \tag{10.36}$$

$$c_m = \left(1 - \frac{1}{2}\frac{c}{h}c_d - \frac{\pi}{3}\frac{A}{h^2}\right)c_{m_{\text{u}}} + \frac{\pi^2}{192}\frac{c^2}{h^2}c_{\ell_{\text{u}}} \tag{10.37}$$

$$\alpha = \alpha_{\text{u}} + \frac{\pi}{24}\frac{c^2}{h^2}\left(c_{m_{\text{u}}} + \frac{1}{4}c_{\ell_{\text{u}}}\right) \tag{10.38}$$

$$Re = \left(1 + \frac{1}{4}\frac{c}{h}c_d + \frac{\pi}{6}\frac{A}{h^2}\right)Re_{\text{u}} \tag{10.39}$$

Equation (10.35) can be solved for c_d explicitly, or on the righthand side $c_d \simeq c_{d_{\text{u}}}$ can be assumed in the usual case where that term is very small.

10.3.2 2D open-jet boundaries

The 2D open-jet wind tunnel considered here still has solid sidewalls which are spanned by the model, these being necessary to obtain 2D flow. However, the top and bottom flow boundaries are free streamlines at a constant ambient pressure p_∞, which will make the overall flow-field different from what it would be in an unbounded flow. Images can again be used to model this effect, as shown in Figure 10.8, but the image signs are different from the solid-wall case. Specifically, the image vortex sheets all have the same sign as those representing the real airfoil, while the source sheets alternate in sign. This gives zero x-velocity perturbations at the jet boundary, and thus to first order makes this boundary a constant-pressure surface.

For concise results we again use the far-field of each image to compute the overall velocity contributions $\Delta u(x)$, $\Delta w(x)$ along the jet centerline, so they can be given by expressions (10.7) and (10.8) which have source, vortex, and doublet components. The kernel functions of these components are now different from

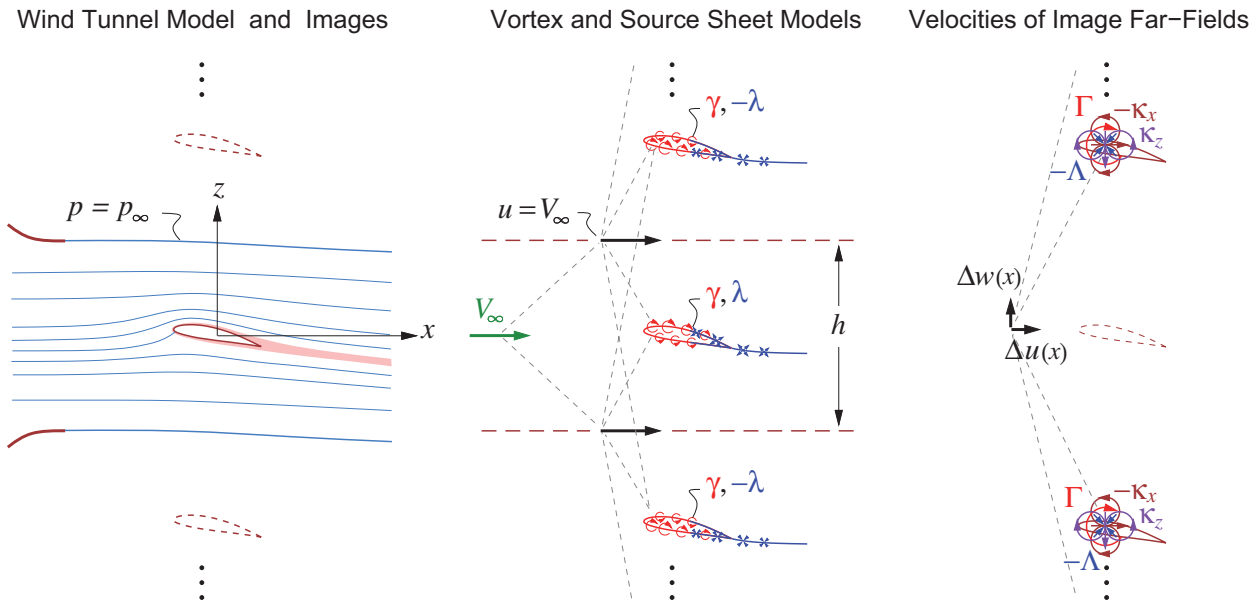

Figure 10.8: 2D airfoil in open-jet tunnel (left) is modeled as an infinite cascade of -/+ images which ensure zero streamwise velocity perturbation and hence constant pressure at the jet boundaries (center). Source, vortex, and doublet far-fields of images produce velocity changes $\Delta u(x)$, $\Delta w(x)$ along the centerline which are absent in unbounded flow (right).

the solid-wall case, and are given as follows.

$$\hat{u}_\Lambda(x) = \frac{1}{\pi}\left[-\frac{X}{1+X^2} + \frac{X}{4+X^2} - \frac{X}{9+X^2} + \ldots\right] \qquad \text{(2D open jet)} \qquad (10.40)$$

$$\hat{w}_\Gamma(x) = \frac{1}{\pi}\left[-\frac{X}{1+X^2} - \frac{X}{4+X^2} - \frac{X}{9+X^2} - \ldots\right] - \frac{1}{2} \qquad \text{(2D open jet)} \qquad (10.41)$$

$$\hat{u}_{\kappa_x}(x) = \frac{1}{\pi}\left[-\frac{1-X^2}{(1+X^2)^2} + \frac{4-X^2}{(4+X^2)^2} - \frac{9-X^2}{(9+X^2)^2} + \ldots\right] \qquad \text{(2D open jet)} \qquad (10.42)$$

$$\hat{w}_{\kappa_z}(x) = \frac{1}{\pi}\left[-\frac{1-X^2}{(1+X^2)^2} - \frac{4-X^2}{(4+X^2)^2} - \frac{9-X^2}{(9+X^2)^2} - \ldots\right] \qquad \text{(2D open jet)} \qquad (10.43)$$

At the model location $X=0$, the sums have the following known analytic results.

$$\hat{u}_\Lambda(0) = 0 \;, \qquad \hat{w}_\Gamma(0) = -\frac{1}{2} \;, \qquad \hat{u}_{\kappa_x}(0) = -\frac{\pi}{12} \;, \qquad \hat{w}_{\kappa_z}(0) = -\frac{\pi}{6} \qquad \text{(2D open jet)} \quad (10.44)$$

Figures 10.9 and 10.10 show the streamline patterns of the resulting $\mathbf{V}_{\text{eff}}(x,z)$ field for each image singularity component. Most of the effects are reversed from the solid-wall case, and the most notable difference is the overall turning of the jet by the model lift.

Volume and wake blockage

In the open-jet wind tunnel, the model's volume (area in 2D) results in the closest images having a negative volume, which produces a local dilation of the $\mathbf{V}_{\text{eff}}$ streamlines. The model's viscous wake displacement results in images with negative displacement, which produces a further spreading. The components of the streamwise velocity perturbation $\Delta u(x)$ along the tunnel axis which are associated with these flow changes are shown in Figure 10.11, which can be compared to Figure 10.6 for the solid-wall case.

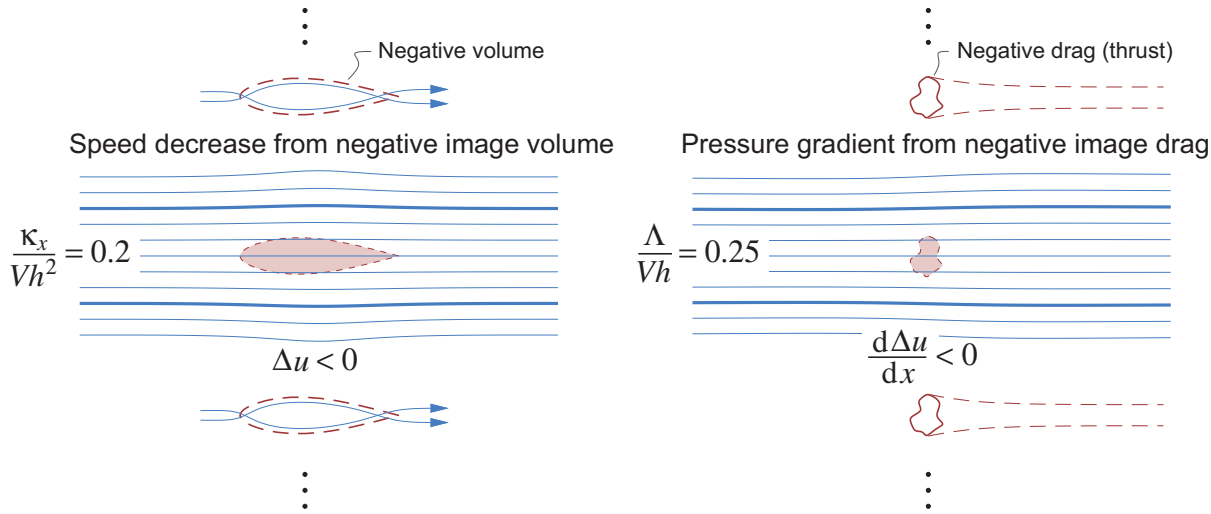

Figure 10.9: Streamlines of the effective freestream $\mathbf{V}_{\text{eff}}(x,z)$ resulting from model images required to model effects of open-jet boundaries. Shown are models with only volume (left) and only profile drag (right). Images have a negative Δu and $\mathrm{d}\Delta u/\mathrm{d}x$ along the jet centerline at the real model location.

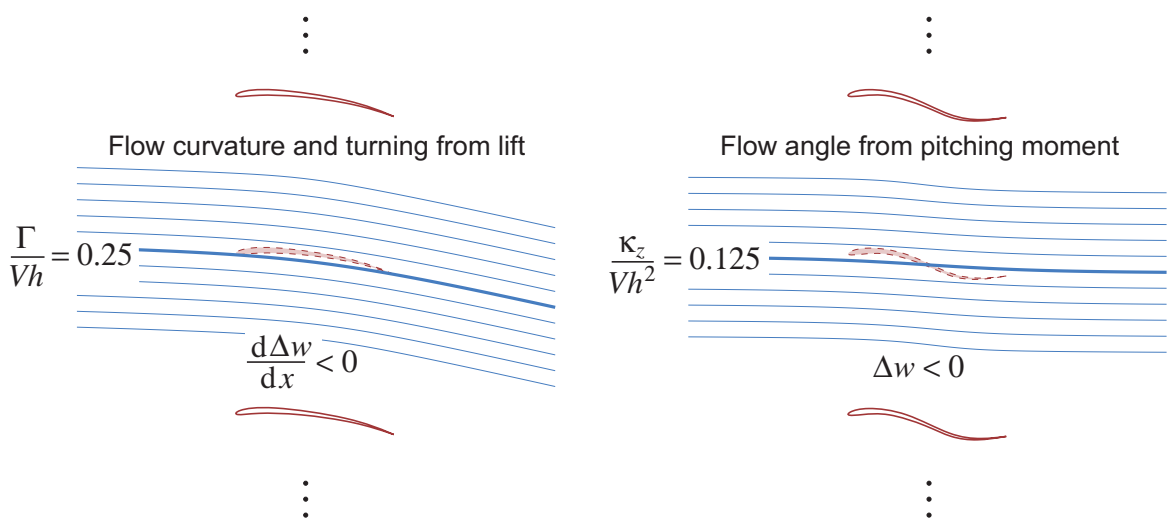

Figure 10.10: Streamlines of the effective freestream $\mathbf{V}_{\text{eff}}(x,z)$ resulting from model images required to model effects of open-jet boundaries. Shown are models with only lift (left) and only pitching moment (right). Images have a negative $\mathrm{d}\Delta w/\mathrm{d}x$ and Δw along the jet centerline at the real model location.

The open-jet boundaries produce a decrease in the velocity and effective dynamic pressure at the model via a negative volume blockage term. Since $\hat{u}_\Lambda(0) = 0$, the wake blockage effect is entirely absent here.

$$\frac{\Delta V}{V_\infty} = \frac{\Delta u(0)}{V_\infty} = -\frac{\pi}{12}\frac{\kappa_x}{V_\infty h^2} = -\frac{\pi}{12}\frac{A}{h^2} \qquad \text{(2D open jet)} \qquad (10.45)$$

$$q_{\text{eff}} \simeq q_\infty\left(1 + 2\frac{\Delta V}{V_\infty}\right) = q_\infty\left(1 - \frac{\pi}{6}\frac{A}{h^2}\right) \qquad (10.46)$$

$$\frac{q_\infty}{q_{\text{eff}}} \simeq 1 + \frac{\pi}{6}\frac{A}{h^2} \qquad \text{(2D open jet)} \qquad (10.47)$$

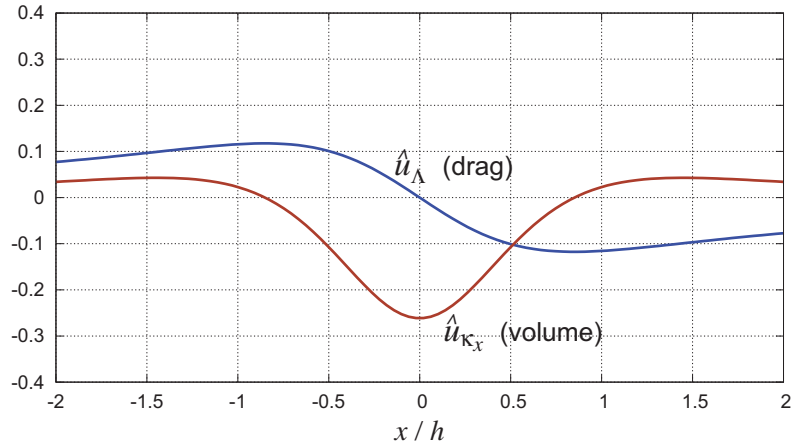

Figure 10.11: Streamwise velocity disturbances along the tunnel jet centerline due to open-jet boundaries, for unit-strength far-field doublet (from model volume) and far-field source (from model drag).

Wake-induced buoyancy

The first images of the model's far-field source are negative, which produces a decelerating effective freestream at the model, as shown in Figure 10.11. Hence the wake-induced buoyancy effect in the open-jet case is opposite in sign and half as large as in the solid-wall case. The resulting buoyancy drag change then also has the opposite sign.

$$\Delta c_{d_{\text{buoy}}} = -\frac{\pi}{12}\frac{A}{h^2}\, c_d \qquad \text{(2D open jet)} \qquad (10.48)$$

Streamline curvature

Compared to the solid-wall case, the curvature effect of lift is opposite in sign, and twice in magnitude.

$$\Delta c_{\ell_{\text{curv}}} = -\frac{\pi^2}{24}\frac{c^2}{h^2}\, c_{\ell_{\text{u}}} \qquad \text{(2D open jet)} \qquad (10.49)$$

$$\Delta c_{m_{\text{curv}}} = \frac{\pi^2}{96}\frac{c^2}{h^2}\, c_{\ell_{\text{u}}} \qquad \text{(2D open jet)} \qquad (10.50)$$

Flow angle

The flow angle change at the model is

$$\begin{aligned}\Delta\alpha_{\text{angle}} &= \frac{\Delta w(0)}{V_\infty} = \frac{\Gamma}{V_\infty h}\,\hat{w}_\Gamma(0) + \frac{\kappa_z}{V_\infty h^2}\,\hat{w}_{\kappa_z}(0) \\ &= -\frac{1}{4}\frac{c}{h}\, c_{\ell_{\text{u}}} - \frac{\pi}{12}\frac{c^2}{h^2}\,(c_{m_{\text{u}}})_{c/2} \qquad \text{(2D open jet)} \qquad (10.51)\end{aligned}$$

where the first lift term did not appear in the solid-wall case, but it dominates here. This term will also tilt the lift vector to create an additional drag component along the tunnel freestream direction.

$$\Delta c_{d_{\text{angle}}} = \frac{1}{4}\frac{c}{h}\, c_{\ell_{\text{u}}}^2 \qquad \text{(2D open jet)} \qquad (10.52)$$

This added drag scales as c_ℓ^2, and thus behaves much like induced drag. It should also be noted that for reasonable c/h ratios this correction will be very large compared to the final corrected drag coefficient. For this reason, measurement of 2D drag in the presence of lift by direct load sensing is likely to be impractical in an open-jet wind tunnel. An alternative method will be considered in Section 10.4.2.

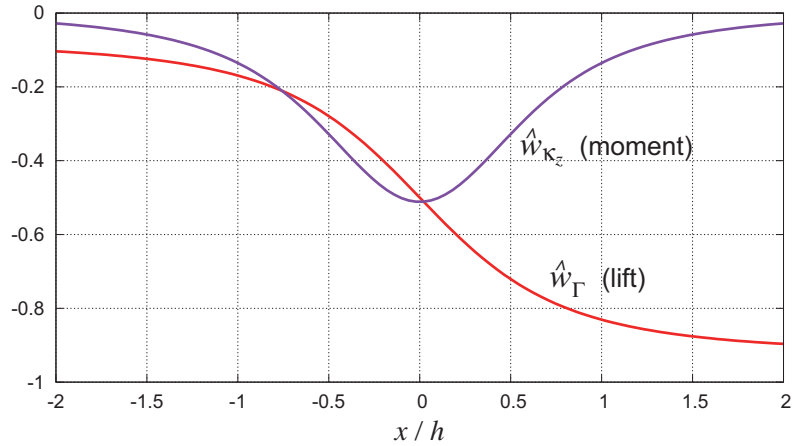

Figure 10.12: Vertical velocity disturbances along the tunnel centerline due to solid walls, for unit-strength far-field vortex (from model lift) and far-field z-doublet (from model pitching moment).

Application of 2D jet corrections

Substituting the changes for the open-jet case into the correction relations (10.30)–(10.32) gives the corrected coefficients and operating parameters in terms of the uncorrected values.

$$c_d = \left(1 + \frac{\pi}{4}\frac{A}{h^2}\right) c_{d\mathrm{u}} - \frac{1}{4}\frac{c}{h} c_{\ell\mathrm{u}}^2 \tag{10.53}$$

$$c_\ell = \left(1 + \frac{\pi}{6}\frac{A}{h^2} + \frac{\pi^2}{24}\frac{c^2}{h^2}\right) c_{\ell\mathrm{u}} \tag{10.54}$$

$$c_m = \left(1 + \frac{\pi}{6}\frac{A}{h^2}\right) c_{m\mathrm{u}} - \frac{\pi^2}{96}\frac{c^2}{h^2} c_{\ell\mathrm{u}} \tag{10.55}$$

$$\alpha = \alpha_\mathrm{u} - \frac{\pi}{12}\frac{c^2}{h^2}\left(c_{m\mathrm{u}} + \frac{1}{4} c_{\ell\mathrm{u}}\right) - \frac{1}{4}\frac{c}{h} c_{\ell\mathrm{u}} \tag{10.56}$$

$$Re = \left(1 - \frac{\pi}{12}\frac{A}{h^2}\right) Re_\mathrm{u} \tag{10.57}$$

10.3.3 3D tunnel images

Figure 10.13 shows a 3D lifting object in a wind tunnel of dimensions $Bh \times h$ (B is the cross-section's aspect ratio). Its far-field is represented by a 3D x-doublet $\mathcal{K}_x$ which models the volume $\mathcal{V}$, a 3D source Σ which models profile drag D_p. These far-field strengths were derived in Section 2.12:

$$\mathcal{K}_x = V_\infty \mathcal{V} \tag{10.58}$$

$$\Sigma = \frac{D_p}{\rho V_\infty} = \frac{1}{2} V_\infty C_{D_p} S_\mathrm{ref} \tag{10.59}$$

In the case of a powered model with engines which produce thrust, D_p would be the <u>net</u> streamwise force on the model excluding induced drag, or "profile drag – thrust." When this is zero, the positive momentum defect of the viscous wakes is canceled by the negative momentum defect of the propulsive jets.

The lifting object trails a vortex wake, which can be lumped into a semi-infinite 2D z-doublet line. Its strength κ_z is obtained from the far-field κ_z definition (2.83), applied to the vortex wake in the Trefftz-Plane yz coordinates by replacing $-\gamma_y x$ with $\gamma_x\, y$. The Trefftz-Plane vorticity relation (5.37) and final lift

result (5.65) are also used.

$$\kappa_z \;=\; \int \gamma_x\, y\, \mathrm{d}s \;=\; \int -\frac{\mathrm{d}(\Delta\varphi)}{\mathrm{d}s}\, y\, \mathrm{d}s \;=\; \int \Delta\varphi\, \mathrm{d}y \;=\; \frac{L}{\rho V_\infty} \;=\; \frac{1}{2} V_\infty\, C_L\, S_{\mathrm{ref}} \tag{10.60}$$

This doublet-line representation assumes that the model's span is small compared to the tunnel dimensions Bh or h. For larger spans it is more realistic to use a horseshoe vortex to represent the model's far-field [67]. But the model-span/tunnel-height ratio then becomes another parameter to consider, and the tunnel corrections also become more complicated. Since the focus here is illustration of the main tunnel-wall effects, the simpler doublet-line representation of the model and vortex wake will be used.

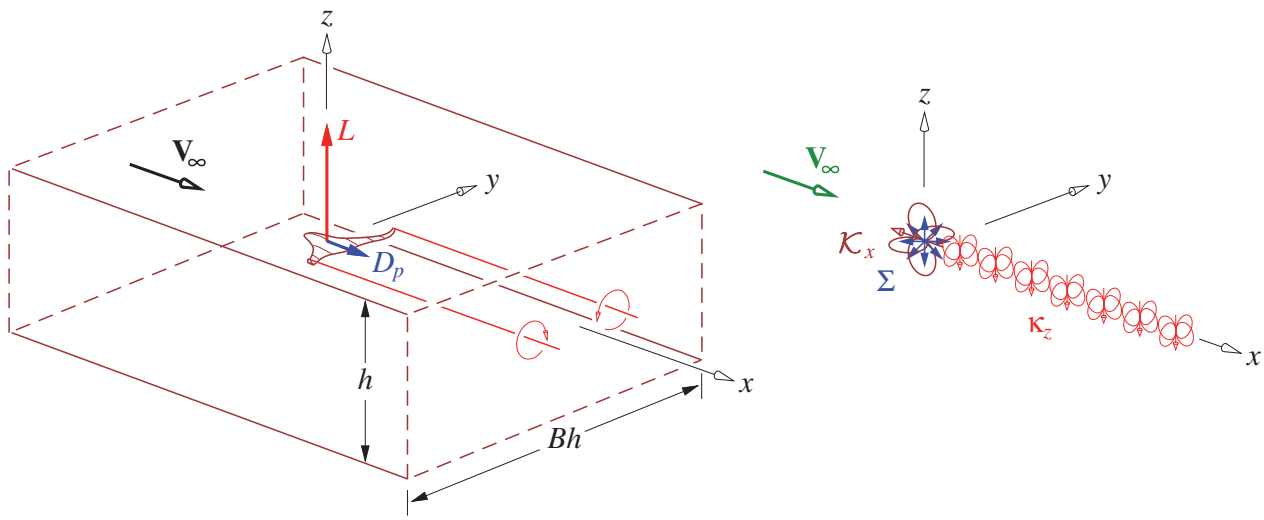

Figure 10.13: 3D lifting object in a wind tunnel of dimensions $Bh \times h$. The far-fields of the object's images are represented by the singularities $\mathcal{K}_x, \Sigma, \kappa_z$, associated by the object's volume, net profile drag, and lift, respectively.

The model's images will produce velocity perturbations along the tunnel axis which in general have the form

$$\Delta u(x\,;B) \;=\; \frac{\Sigma}{h^2}\,\hat{u}_\Sigma \;+\; \frac{\mathcal{K}_x}{h^3}\,\hat{u}_{\mathcal{K}_x} \tag{10.61}$$

$$\Delta w(x\,;B) \;=\; \frac{\kappa_z}{h^2}\,\hat{w}_{\kappa_z} \tag{10.62}$$

$$\frac{\mathrm{d}\Delta u}{\mathrm{d}x}(x\,;B) \;=\; \frac{\Sigma}{h^3}\,\frac{\mathrm{d}\hat{u}_\Sigma}{\mathrm{d}X} \;=\; \frac{\Sigma}{h^3}\,\hat{u}_{\mathcal{K}_x} \tag{10.63}$$

$$\frac{\mathrm{d}\Delta w}{\mathrm{d}x}(x\,;B) \;=\; \frac{\kappa_z}{h^3}\,\frac{\mathrm{d}\hat{w}_{\kappa_z}}{\mathrm{d}X} \tag{10.64}$$

where the unit-strength velocity kernels $\hat{u}_\Sigma, \hat{u}_{\mathcal{K}_x}, \hat{w}_{\kappa_z}$ are the sums over all the images as in the 2D case. The streamwise derivative of Δw is also summed to obtain flow curvature corrections. It's convenient to first define the unit-strength x and z velocities $\hat{U}$ and $\hat{W}$ for each image singularity.

$$\hat{U}_\Sigma(x,y,z) \;=\; \frac{1}{4\pi}\,\frac{X}{[X^2+Y^2+Z^2]^{3/2}} \tag{10.65}$$

$$\hat{U}_{\mathcal{K}_x}(x,y,z) \;=\; \frac{1}{4\pi}\,\frac{Y^2+Z^2-2X^2}{[X^2+Y^2+Z^2]^{5/2}} \tag{10.66}$$

$$\hat{W}_{\kappa_z}(x,y,z) \;=\; \frac{1}{4\pi}\,\frac{1}{Y^2+Z^2}\left\{\frac{Y^2-Z^2}{Y^2+Z^2}\left(\frac{X}{[X^2+Y^2+Z^2]^{1/2}}+1\right) - \frac{XZ^2}{[X^2+Y^2+Z^2]^{3/2}}\right\} \tag{10.67}$$

$$\frac{\mathrm{d}\hat{W}_{\kappa_z}}{\mathrm{d}X}(x,y,z) \;=\; \frac{1}{4\pi}\,\frac{1}{Y^2+Z^2}\left\{\frac{Y^2-Z^2}{[X^2+Y^2+Z^2]^{3/2}} - Z^2\,\frac{Y^2+Z^2-2X^2}{[X^2+Y^2+Z^2]^{5/2}}\right\} \tag{10.68}$$

10.3.4 3D solid-wall boundaries

For the tunnel cross-section and model shown in Figure 10.13, the flow tangency requirement on all four solid walls is imposed by the array of images shown in Figure 10.14. As in the 2D case, the images produce axial and vertical velocity perturbations along the tunnel centerline, given by summing over all the images in the yz plane. The j index sums along y, and the k index sums along z.

$$\hat{u}_{\mathcal{K}_x}(x\,;B) = \sum_{j=-\infty}^{\infty}\sum_{k=-\infty}^{\infty} \hat{U}_{\mathcal{K}_x}(X, jB, k) \qquad \text{(3D solid walls)} \quad (10.69)$$

$$\hat{u}_{\Sigma}(x\,;B) = \sum_{j=-\infty}^{\infty}\sum_{k=-\infty}^{\infty} \hat{U}_{\Sigma}(X, jB, k) + \frac{1}{2B} \qquad \text{(3D solid walls)} \quad (10.70)$$

$$\hat{w}_{\kappa_z}(x\,;B) = \sum_{j=-\infty}^{\infty}\sum_{k=-\infty}^{\infty} (-1)^k\, \hat{W}_{\kappa_z}(X, jB, k) \qquad \text{(3D solid walls)} \quad (10.71)$$

$$\frac{\mathrm{d}\hat{w}_{\kappa_z}}{\mathrm{d}X}(x\,;B) = \sum_{j=-\infty}^{\infty}\sum_{k=-\infty}^{\infty} (-1)^k\, \frac{\mathrm{d}\hat{W}_{\kappa_z}}{\mathrm{d}X}(X, jB, k) \qquad \text{(3D solid walls)} \quad (10.72)$$

The $j=0, k=0$ self-influence term is omitted from the above double sums. Along the tunnel axis, these velocity perturbations look qualitatively similar to those for the 2D case shown in Figures 10.6 and 10.7. Their values at the $x=0$ model location are plotted versus the section aspect ratio B in Figure 10.15.

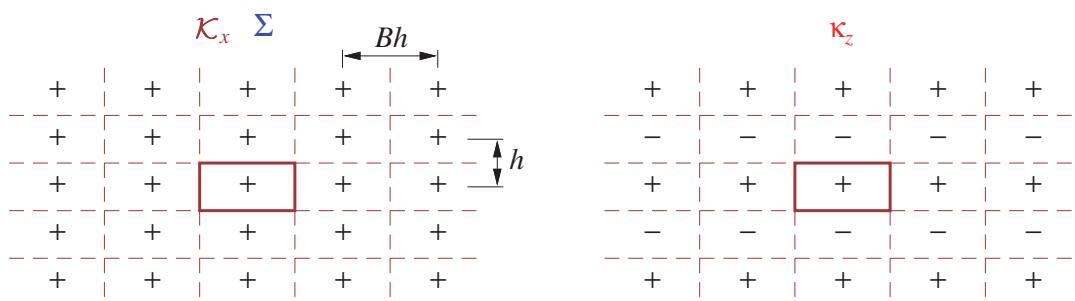

Figure 10.14: Image arrays around a 3D tunnel cross section, needed to produce zero normal velocity at the walls (solid rectangle), Left figure gives signs of the image 3D x-doublet and source strengths. Right figure gives signs of the image 2D z-doublet strengths.

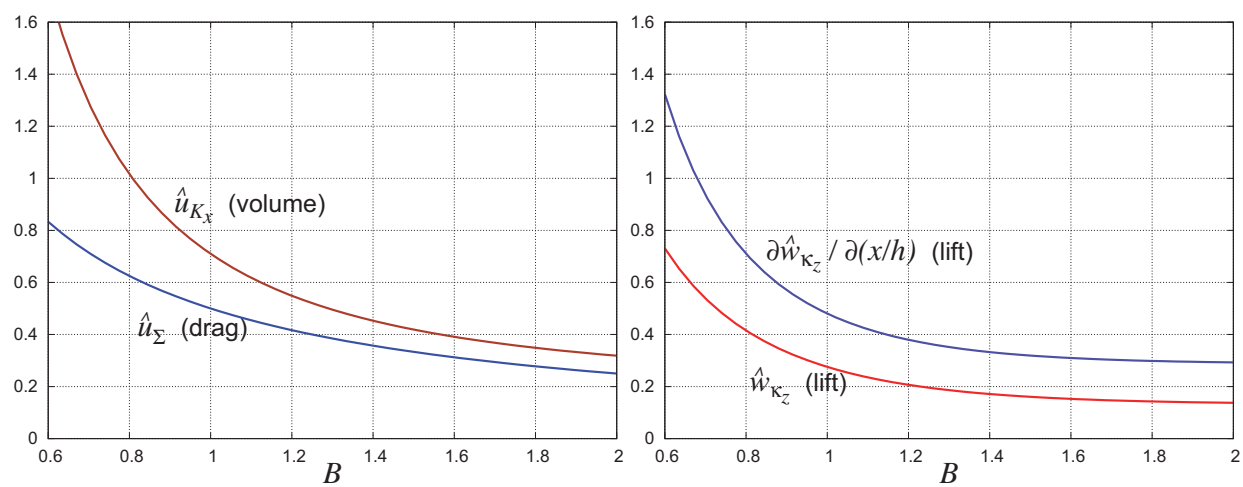

Figure 10.15: 3D velocity disturbances at the model location $x=0$ versus tunnel section aspect ratio, for a solid-wall tunnel.

10.3.5 3D open-jet boundaries

The constant pressure or $\Delta u = 0$ requirement on the open-jet boundary is obtained with the image arrays shown in Figure 10.16. The perturbation velocities are given by the following sums over the images.

$$\hat{u}_{\mathcal{K}_x}(x\,;B) = \sum_{j=-\infty}^{\infty}\sum_{k=-\infty}^{\infty}(-1)^{(j+k)}\hat{U}_{\mathcal{K}_x}(X,jB,k) \qquad \text{(3D open jet)} \qquad (10.73)$$

$$\hat{u}_{\Sigma}(x\,;B) = \sum_{j=-\infty}^{\infty}\sum_{k=-\infty}^{\infty}(-1)^{(j+k)}\hat{U}_{\Sigma}(X,jB,k) \qquad \text{(3D open jet)} \qquad (10.74)$$

$$\hat{w}_{\kappa_z}(x\,;B) = \sum_{j=-\infty}^{\infty}\sum_{k=-\infty}^{\infty}(-1)^{j}\hat{W}_{\kappa_z}(X,jB,k) \qquad \text{(3D open jet)} \qquad (10.75)$$

$$\frac{\mathrm{d}\hat{w}_{\kappa_z}}{\mathrm{d}X}(x\,;B) = \sum_{j=-\infty}^{\infty}\sum_{k=-\infty}^{\infty}(-1)^{j}\frac{\mathrm{d}\hat{W}_{\kappa_z}}{\mathrm{d}X}(X,jB,k) \qquad \text{(3D open jet)} \qquad (10.76)$$

Again, the $j = 0, k = 0$ self-influence term is omitted from these double sums. The velocity values at the $x = 0$ model location are plotted versus the section aspect ratio B in Figure 10.17.

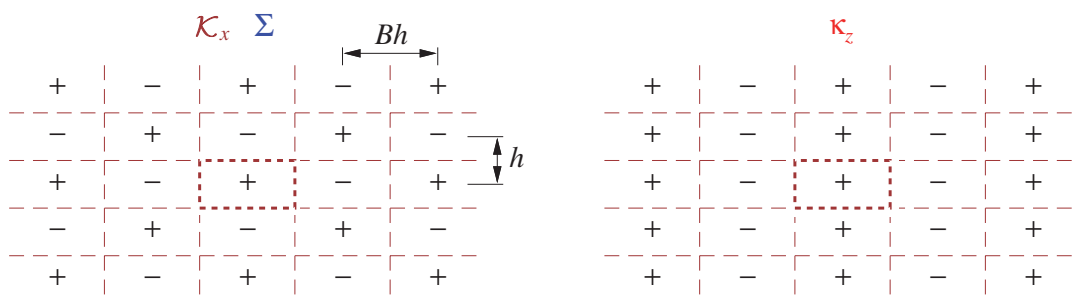

Figure 10.16: Image arrays around a 3D tunnel cross section needed to produce $\Delta u = 0$ at the jet boundary (dotted), thus representing an open jet. Left figure gives signs of the image 3D x-doublet and source strengths. Right figure gives signs of the image 2D z-doublet strengths.

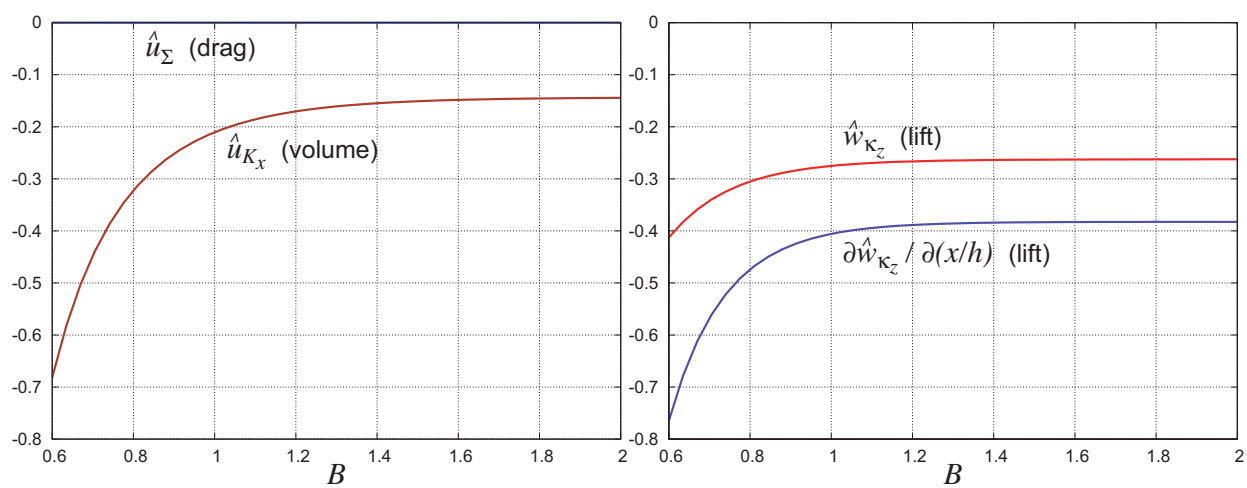

Figure 10.17: 3D velocity disturbances at the model location $x = 0$ versus tunnel section aspect ratio, for an open-jet tunnel.

The 3D image velocities derived above are the starting point for application to actual tunnel corrections as described next. The corrections for solid-wall and open-jet tunnels have the same form, except that the magnitudes and signs are different.

Volume and wake blockage

The blockage effects are essentially the same as in the 2D case, as given by (10.18) and (10.19).

$$\frac{\Delta V}{V_\infty} = \frac{\Delta u(0)}{V_\infty} = \frac{\Sigma}{V_\infty h^2}\,\hat{u}_\Sigma(0) + \frac{\mathcal{K}_x}{V_\infty h^3}\,\hat{u}_{\mathcal{K}_x}(0) = \frac{S_{\rm ref}}{h^2}\,\frac{C_{D_p}}{2}\,\hat{u}_\Sigma(0) + \frac{\mathcal{V}}{h^3}\,\hat{u}_{\mathcal{K}_x}(0) \tag{10.77}$$

$$\frac{q_\infty}{q_{\rm eff}} \simeq 1 - 2\frac{\Delta V}{V_\infty} = 1 - \frac{S_{\rm ref}}{h^2}\,C_{D_p}\,\hat{u}_\Sigma(0) - 2\frac{\mathcal{V}}{h^3}\,\hat{u}_{\mathcal{K}_x}(0) \tag{10.78}$$

The 3D far-field models (2.110),(2.113) have been used to express Σ and $\mathcal{K}_x$ in term of the model's profile drag coefficient and volume. The velocity kernel factors $\hat{u}_\Sigma(0)$ and $\hat{u}_{\mathcal{K}_x}(0)$ depend on the tunnel cross-section aspect ratio B, and can be obtained from Figure 10.15 for a solid-wall tunnel, or from Figure 10.17 for an open-jet tunnel.

Wake-induced buoyancy

The streamwise acceleration from the wake displacement is also computed the same way as in 2D.

$$\frac{{\rm d}V_{\rm eff}}{{\rm d}x} = \frac{{\rm d}\Delta V}{{\rm d}x} = \frac{\Lambda}{h^3}\,\hat{u}_{\mathcal{K}_x}(0) = \frac{V_\infty}{h}\,\frac{S_{\rm ref}}{h^2}\,\frac{C_{D_p}}{2}\,\hat{u}_{\mathcal{K}_x}(0) \tag{10.79}$$

The resulting streamwise pressure gradient acts on the body's volume $\mathcal{V}$ to produce an additional buoyancy drag which must be removed from the measured drag.

$$\Delta C_{D_{\rm buoy}} = \frac{1}{\frac{1}{2}\rho V_\infty^2\,S_{\rm ref}}\,\rho V_\infty \frac{{\rm d}V_{\rm eff}}{{\rm d}x}\,\mathcal{V} = \frac{\mathcal{V}}{h^3}\,C_{D_p}\,\hat{u}_{\mathcal{K}_x}(0) \tag{10.80}$$

Flow angle

The vertical velocity Δw at the model will produce an angle correction as in the 2D case.

$$\Delta\alpha_{\rm angle} = \frac{\Delta w(0)}{V_\infty} = \frac{\kappa_z}{V_\infty h^2}\,\hat{w}_{\kappa_z}(0) = \frac{1}{2}\,\frac{S_{\rm ref}}{h^2}\,C_{L\rm u}\,\hat{w}_{\kappa_z}(0) \tag{10.81}$$

This angle will also tilt the lift vector and produce a drag correction which must be removed from the measured drag.

$$\Delta C_{D_{\rm angle}} = -C_{L\rm u}\,\Delta_{\rm angle} = -\frac{1}{2}\,\frac{S_{\rm ref}}{h^2}\,C_{L\rm u}^2\,\hat{w}_{\kappa_z}(0) \tag{10.82}$$

This drag correction scales as C_L^2 like induced drag, and is in fact equivalent to "ground effect" for the solid-wall case. For the open-jet case the correction has the opposite sign.

Streamline curvature

The flow curvature is given by the streamwise derivative of Δw as in the 2D case.

$$\chi = \frac{\kappa_z}{V_\infty h^3}\,\frac{{\rm d}\hat{w}_{\kappa_z}}{{\rm d}X}(0) = \frac{1}{2h}C_{L\rm u}\,\frac{S_{\rm ref}}{h^2}\,\frac{{\rm d}\hat{w}_{\kappa_z}}{{\rm d}X}(0) \tag{10.83}$$

The effect on the lift and moment coefficients depends on the model configuration, since different components will see different flow angles in the presence of the curved flow. However, we note that the flow angle distribution due to the streamline curvature is the same as that generated by an equivalent pitch rate

$q_{\rm curv} = {\rm d}\Delta w/{\rm d}x = V_\infty \chi$. Hence, it is reasonable and convenient to quantify the flow curvature effect in terms of the aircraft's pitch-rate stability derivatives C_{L_q}, C_{m_q}, treated in Chapter 6.

$$\Delta C_{L_{\rm curv}} = C_{L_q} \frac{\Delta q\, c_{\rm ref}}{2V_\infty} = C_{L_q} \frac{\chi\, c_{\rm ref}}{2} = \frac{1}{4} C_{L_{\rm u}}\, C_{L_q} \frac{S_{\rm ref} c_{\rm ref}}{h^3} \frac{{\rm d}\hat{w}_{\kappa_z}}{{\rm d}X}(0) \tag{10.84}$$

$$\Delta C_{m_{\rm curv}} = C_{m_q} \frac{\Delta q\, c_{\rm ref}}{2V_\infty} = C_{m_q} \frac{\chi\, c_{\rm ref}}{2} = \frac{1}{4} C_{L_{\rm u}}\, C_{m_q} \frac{S_{\rm ref} c_{\rm ref}}{h^3} \frac{{\rm d}\hat{w}_{\kappa_z}}{{\rm d}X}(0) \tag{10.85}$$

The change in lift coefficient will also produce a change in drag coefficient. Here we will assume that only the induced drag component is affected.

$$\Delta C_{D_{\rm curv}} = \frac{{\rm d}C_{D_i}}{{\rm d}C_L} \Delta C_{L_{\rm curv}} = \frac{2C_{L_{\rm u}}}{\pi\, AR\, e} \Delta C_{L_{\rm curv}} \tag{10.86}$$

Application of 3D tunnel corrections

The force and moment coefficients are corrected for the tunnel boundary velocity perturbations in the same basic manner as in the 2D case.

$$C_D = \frac{q_\infty}{q_{\rm eff}} \left(C_{D_{\rm u}} - \Delta C_{D_{\rm curv}} - \Delta C_{D_{\rm buoy}} - \Delta C_{D_{\rm angle}} \right) \tag{10.87}$$

$$C_L = \frac{q_\infty}{q_{\rm eff}} \left(C_{L_{\rm u}} - \Delta C_{L_{\rm curv}} \right) \tag{10.88}$$

$$C_m = \frac{q_\infty}{q_{\rm eff}} \left(C_{m_{\rm u}} - \Delta C_{m_{\rm curv}} \right) \tag{10.89}$$

The angle of attack and Reynolds number which the model effectively sees are also obtained by correcting for the perturbed flow angle and flow speed.

$$\alpha = \alpha_{\rm u} + \Delta\alpha_{\rm angle} \tag{10.90}$$

$$Re = \frac{V_{\rm eff}}{V_\infty} Re_{\rm u} \tag{10.91}$$

Substituting the various changes into the correction relations derived from the solid-wall image models and dropping the higher-order terms, gives the corrected coefficients in terms of the uncorrected values.

$$\begin{aligned} C_D = {} & \left(1 - \frac{S_{\rm ref}}{h^2} C_{D_p}\, \hat{u}_\Sigma(0) - 2\frac{\mathcal{V}}{h^3}\, \hat{u}_{\mathcal{K}_x}(0)\right) C_{D_{\rm u}} - \frac{1}{2} \frac{S_{\rm ref} c_{\rm ref}}{h^3} C_{L_q} \frac{C_{L_{\rm u}}^2}{\pi AR e} \frac{{\rm d}\hat{w}_{\kappa_z}}{{\rm d}X}(0) \\ & - \frac{\mathcal{V}}{h^3} C_{D_p}\, \hat{u}_{\mathcal{K}_x}(0) + \frac{1}{2} \frac{S_{\rm ref}}{h^2} C_{L_{\rm u}}^2\, \hat{w}_{\kappa_z}(0) \end{aligned} \tag{10.92}$$

$$C_L = \left(1 - \frac{S_{\rm ref}}{h^2} C_{D_p}\, \hat{u}_\Sigma(0) - 2\frac{\mathcal{V}}{h^3}\, \hat{u}_{\mathcal{K}_x}(0)\right) C_{L_{\rm u}} - \frac{1}{4} \frac{S_{\rm ref} c_{\rm ref}}{h^3} C_{L_q}\, C_{L_{\rm u}} \frac{{\rm d}\hat{w}_{\kappa_z}}{{\rm d}X}(0) \tag{10.93}$$

$$C_m = \left(1 - \frac{S_{\rm ref}}{h^2} C_{D_p}\, \hat{u}_\Sigma(0) - 2\frac{\mathcal{V}}{h^3}\, \hat{u}_{\mathcal{K}_x}(0)\right) C_{m_{\rm u}} - \frac{1}{4} \frac{S_{\rm ref} c_{\rm ref}}{h^3} C_{m_q}\, C_{L_{\rm u}} \frac{{\rm d}\hat{w}_{\kappa_z}}{{\rm d}X}(0) \tag{10.94}$$

$$\alpha = \alpha_{\rm u} + \frac{1}{2} \frac{S_{\rm ref}}{h^2} C_{L_{\rm u}}\, \hat{w}_{\kappa_z}(0) \tag{10.95}$$

$$Re = \left(1 + \frac{1}{2} \frac{S_{\rm ref}}{h^2} C_{D_p}\, \hat{u}_\Sigma(0) + \frac{\mathcal{V}}{h^3}\, \hat{u}_{\mathcal{K}_x}(0)\right) Re_{\rm u} \tag{10.96}$$

Note that application of these corrections requires knowing the profile drag and induced drag components of the total drag.

10.4 2D tunnel drag measurements

10.4.1 Flow two-dimensionality requirements

Valid 2D wind tunnel experiments, regardless of the measurement techniques used, in general require that the flow is actually two-dimensional to a sufficient accuracy. The model must span the entire test section, since any significant gaps between the model wingtips and the tunnel walls will result in the generation of trailing vorticity and unwanted induced drag. Unfortunately, the model wingtips will adversely interact with the tunnel-wall boundary layers, which will usually add spurious pressure drag to the model. The wall boundary layer fluid can also flow spanwise onto the model and disturb the model's ideally 2D boundary layers. To avoid these problems, the tunnel wall boundary layers just ahead of the model ideally should be removed with local wall suction, or energized with tangential blowing. Also, the model aspect ratio b/c can be made large to reduce the effect of the ends. Large tunnel height to model chord ratios h/c are also desirable to minimize the tunnel-wall effects and the associated corrections, which will be discussed later.

Assuming that sufficient precautions are taken to ensure good flow two-dimensionality, the drag of a 2D model with negligible lift can be satisfactorily measured with a load sensor. However, measurement of the drag of a 2D lifting airfoil using a load sensor is extraordinarily difficult. Because a 2D airfoil can easily have $L'/D' \simeq 100$ or more, the drag error from load-sensor misalignment as given by equation (10.4) becomes overwhelming. For example, a misalignment error of $\theta_{\text{err}} = 1^\circ$ and $L'/D' = 100$ will give 175% error in the measured drag, rendering the experimental results meaningless. Furthermore, even small drag additions from the model/wall junction interaction can overwhelm the model's small drag. In practice, the direct drag force measurement approach is simply not suitable for 2D lifting airfoils. A good alternative is the wake momentum drag measurement technique, which is described next.

10.4.2 Wake momentum drag measurement

This method seeks to measure the profile drag indirectly, via the far-field momentum defect of the airfoil's wake which is defined in terms of the wake velocity profile $u(z)$. This in turn is obtained from the wake total pressure profile $p_o(z)$ measured via a *wake rake* (or *drag rake*), which is an array of pitot tubes with spacing Δz, as shown in Figure 10.18. The local static pressure p_{e} at the rake location is also measured via a static probe. This static pressure is typically assumed to be constant across the wake.

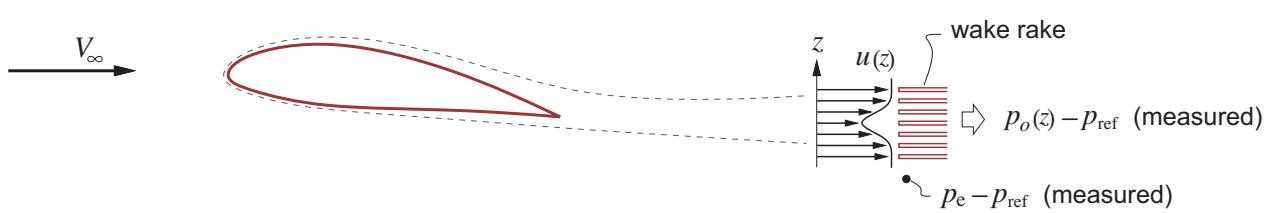

Figure 10.18: Wake measurement via a wake rake for determination of 2D profile drag.

Most pressure sensors report a difference between two pressures, with one of these being some convenient chosen reference pressure p_{ref} supplied to the sensor. Typically this is either p_∞ or p_{o_∞}, or some other comparable pressure. Regardless, the reported data is actually $p_o(z) - p_{\text{ref}}$ and $p_{\text{e}} - p_{\text{ref}}$.

The wake velocity profile is computed from the rake pressure data and the definition of total pressure.

$$u(z) = \sqrt{\frac{2[p_o(z) - p_{\text{e}}]}{\rho}} = \sqrt{\frac{2[(p_o(z) - p_{\text{ref}}) - (p_{\text{e}} - p_{\text{ref}})]}{\rho}} \tag{10.97}$$

$$u_{\text{e}} = u(z_{\text{e}}) \tag{10.98}$$

The displacement and momentum thicknesses and corresponding shape parameter are then computed by numerical integration over the $i = 1 \ldots N$ rake tubes.

$$\delta^* = \int \left(1 - \frac{u}{u_\mathrm{e}}\right) \mathrm{d}z \simeq \sum_{i=1}^{N} \left(1 - \frac{u_i}{u_\mathrm{e}}\right) \Delta z \tag{10.99}$$

$$\theta = \int \left(1 - \frac{u}{u_\mathrm{e}}\right) \frac{u}{u_\mathrm{e}} \, \mathrm{d}z \simeq \sum_{i=1}^{N} \left(1 - \frac{u_i}{u_\mathrm{e}}\right) \frac{u_i}{u_\mathrm{e}} \, \Delta z \tag{10.100}$$

$$H = \frac{\delta^*}{\theta} \tag{10.101}$$

From the far-field force analysis given in Appendix C, the profile drag/span is the momentum defect far downstream in the wake.

$$D' = P_\infty = \rho V_\infty^2 \, \theta_\infty \tag{10.102}$$

$$c_d \equiv \frac{D'}{\frac{1}{2}\rho V_\infty^2 \, c} = 2 \, \frac{\theta_\infty}{c} \tag{10.103}$$

The wake reaches its final value θ_∞ downstream of the airfoil's static pressure near-field, where $p_\mathrm{e} = p_\infty$ and correspondingly $u_\mathrm{e} = V_\infty$. Hence, the rake should ideally be placed far downstream, well away from the airfoil's near-field. One practical difficulty with this is that the wake spreads rapidly downstream, and the velocity defect profile $u_\mathrm{e} - u(z)$ becomes small and is difficult to measure accurately. So for measurement accuracy of the velocity profile it's desirable to measure the wake close behind the airfoil where $p_\mathrm{e} \neq p_\infty$ in general.

This conflict is resolved by measuring the wake fairly close behind the airfoil, and then extrapolating the results to far-downstream conditions using the Squire-Young formula (4.118) which was derived and discussed in Section 4.13.3, and is restated as follows for convenience.

$$\theta_\infty = \theta(\bar{x}) \left(\frac{u_\mathrm{e}(\bar{x})}{V_\infty}\right)^{(H_\mathrm{avg}+2)} \tag{10.104}$$

$$\text{where} \qquad H_\mathrm{avg} = \frac{H(\bar{x}) + 1}{2} \tag{10.105}$$

Coincidentally, this relation was originally developed by Squire and Young [34] for the calculation of profile drag by extrapolation from the trailing edge quantities, thus avoiding shear layer calculations into the wake. Here, it is used to avoid the need to measure the wake too far downstream.